Dimensionality Reduction and Classification of Hyperspectral Image via Multistructure Unified Discriminative Embedding

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***Abstract*— Graph can achieve good performance to extract the low-dimensional features of hyperspectral image (HSI). However, the present graph-based methods just consider the individual information of each sample in a certain characteristic, which is very difficult to represent the intrinsic properties of HSI for the complex imaging condition. To better represent the low-dimensional features of HSI, we propose a multistructure unified discriminative embedding (MUDE) method, which considers the neighborhood, tangential, and statistical properties of each sample in HSI to achieve the complementarity of different characteristics. In MUDE, we design the intraclass and interclass neighborhood structure graphs with the local reconstruction structure of each sample; meanwhile, we also utilize the adaptive tangential affine combination structure to construct the intraclass and interclass tangential structure graphs. To further enhance the discriminating performance between different classes, we consider the influence of the statistical distribution difference for each sample to develop an interclass Gaussian weighted scatter model. Then, an embedding objective function is constructed to enhance the intraclass compactness and the interclass separability and obtain more discriminative features for HSI classification. Experiments on three real HSI datasets show that the proposed method can make full use of the structure information of each sample in different characteristics to achieve the complementarity of different features and improve the classification performance of HSI compared with the state-of-the-art methods.**

***Index Terms*— Dimensionality reduction (DR), discriminative learning, graph learning, hyperspectral image (HSI), tangent space.**

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I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of channels captured by imaging spectrometer under narrow and continuous electromagnetic spectrums covering from visible light to infrared light [1]–[3]. In HSI intelligent interpretation, each pixel is considered as a high-dimensional vector, which provides rich characteristic information of target objects [4]–[6]. Since different target objects contain different characteristic information, the type of each pixel can be recognized according to the differences of high-dimensional vectors [7], [8]. However, there is a large amount of redundant information in high-dimensional HSI data, which will have an adverse effect on the discriminant performance of features for HSI classification task [9]–[11].

To enhance the discriminant performance of features, a series of dimensionality reduction (DR) algorithms has been designed to reduce some redundant information in HSI, while they can preserve more significant valuable features [12]–[14]. Many classic DR methods are based on statistical theory, such as principal component analysis (PCA) [15], linear discriminant analysis (LDA) [16], and nonparametric weighted feature extraction (NWFE) [17], which aims to find a new data projection space with a certain statistical criterion. Meanwhile, a series of manifold learning methods was developed with the local structure of data [18]. Unlike the statistics-based methods to learn the statistical features of data, these manifold-based methods can reveal the intrinsic geometry structures of data [19], [20], which has been widely used in the field of HSI processing, including locally linear embedding (LLE) [21], Laplacian eigenmaps (LEs) [22], and their extended algorithms such as neighbor preserving embedding (NPE) [23] and local preserving projection (LPP) [24]. To better understand and analyze these DR algorithms, a graph embedding (GE) framework has been proposed to unify these existing methods [25], [26]. In the GE framework, with some specific statistical or geometric properties, it constructs a similarity graph to preserve the homogeneous relationship of data that are beneficial to the discriminant performance of embedding features; meanwhile, a penalty graph is also designed to suppress the heterogeneous relationship of data that are useless to the low-dimensional discriminant features [27], [28]. Based on this framework, a lot of new algorithms have been developed, such as marginal Fisher analysis (MFA) [29] and local geometric structure Fisher analysis (LGSFA) [30]. For the

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above-mentioned GE methods, they often rely on the Euclidean distance to explore the intrinsic geometric relationship of data with a fixed neighborhood size, which is very sensitive to noise, affecting the accuracy of graph construction and decreasing the discriminant performance of embedding features [31], [32].

To effectively explore the intrinsic relationship of data, sparse representation (SR) has been introduced into the field of machine learning that can adaptively reveal the similar relationship of data via L0 or L1 norm [33]–[35]. SR uses a dictionary to robustly represent each sample, in which the representation coefficients are sparse to adaptively reveal the intrinsic relationship of data [36], [37]. Therefore, some SR-based methods have also been developed to reduce the dimensionality of high-dimensional data. Sparsity preserving projection (SPP) [38] and sparsity preserving graph embedding (SPGE) [39] are two classic unsupervised SR algorithms to extract features by preserving the sparse reconstruction relationship of data. To make full use of the prior information of data, many supervised SR methods were proposed to reduce the dimensionality of high-dimensional data, such as sparse graph-based discriminant analysis (SGDA) [39] and discriminative learning by SR projections (DLSP) [40]. These supervised methods use the sparse coefficients and the class label to construct sparse graphs and adaptively extract the intrinsic sparsity of data. However, SR just reveals the local property of samples. To better represent the global property of all the samples, sparse and low-rank graph-based discriminant analysis (SLGDA) [41] was proposed by using the combined sparse and low-rank representations to construct a sparse low-rank graph, which improves the robustness of embedding features. To consider the spatial–spectral properties of HSI, a tensor is used to construct the sparse and low-rank graph and propose a patch tensor-based sparse and low-rank graph (PT-SLG) method [42] and a tensor-based low-rank graph with multimanifold regularization (T-LGMR) method [43]. To explore the collaborative property between samples, collaborative representation was introduced to construct a collaborative graph and propose the collaborative graph-based discriminant analysis (CGDA) method [44], [45]. To further consider the geometric structures of data in SR, a sparse manifold representation was developed by combining the sparse and manifold properties of data. Based on the sparse manifold representation, some methods utilized the sparse manifold coefficients to design different graphs and propose sparse discriminant manifold embedding (SDME) [46], local manifold-based sparse discriminant learning (LMSDL) [47], and local constraint-based sparse manifold hypergraph learning (LCSMHL) [48].

To summarize the above GE algorithms, these methods just use the individual information of each sample to construct graphs, which neglects the structure information of each sample. For complex HSI, the methods are very difficult to effectively reveal the intrinsic structure properties with the individual information because the complex imaging condition generates a large difference in homogeneous spectra [49]. In addition, for the structure complexity of HSI, only considering one characteristic information, it is very difficult to comprehensively and effectively represent the intrinsic features. Therefore, to reduce the homogeneous spectra difference and better represent the intrinsic features of HSI, we consider the structure information of each sample to construct a graph in this article and propose multistructure unified discriminative embedding (MUDE). In MUDE, we use the neighborhood, tangential, and statistical properties to represent the structures of each sample in HSI, which can achieve the complementarity of different structures of HSI to effectively extract more discriminative features. The proposed MUDE enhances the intraclass compactness and the interclass separability for subsequent classification task. The main contributions of this article include the following.

1) For the previous graph learning, it just represents the individual similar relationship between two samples. However, in complex HSI, each sample and its neighborhood can be considered as an approximately local linear structure. We utilize the local reconstruction structure of each sample to construct an intraclass neighborhood structure graph and an interclass neighborhood structure graph, which makes full use of the neighborhood information to better represent the intrinsic characteristic of HSI and enhance the discriminant performance of embedding features.
2) In HSI, each sample contains the radial direction information and the tangential information. The neighborhood graph represents the radial direction properties of HSI. To achieve the complementarity with the radial direction information, we consider the tangential properties of HSI and propose a tangential graph to learn the tangential information of each sample in HSI. Meanwhile, we consider the tangential structure of each sample in tangent space to construct an intraclass tangential structure graph and an interclass tangential structure graph. Here, we use a tangential affine combination based on SR to adaptively reveal the tangential structure of each sample.
3) For the influence of the complex imaging condition and the reflection difference of spectra at different bands, the spectra have serious distortion in HSI, making it very difficult to represent the distribution of HSI data by a unified model. To further reduce the influence of data distribution and enhance the discriminating performance of different classes, we consider the statistical properties of HSI to construct an interclass weighted scatter model. Here, we use a Gaussian distribution weight to improve the scatter of the embedding features of different classes.
4) Based on the neighbor, tangential, and statistical properties of HSI, we combine this different characteristic information of HSI to design a low-dimensional embedding objective function, which compacts the intraclass samples as close as possible and separates the interclass samples as far as possible. Then, we can obtain the discriminant features in low-dimensional space, which achieves the complementarity of different information and enhances the discriminant performance of embedding features for HSI classification.

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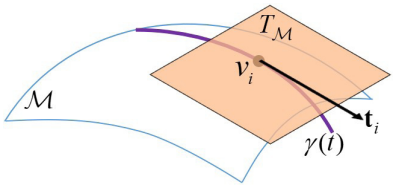


Fig. 1. Example for tangent space.

The organization of the rest of this article includes the following. Section II introduces some basic definitions and related algorithms. The proposed MUDE method is detailedly described in Section III. Section IV conducts some experiments on three real HSI datasets to demonstrate the effectiveness of MUDE. In Section V, we analyze some conclusions of this article and suggestions for future work.

II. Related Works

We assume that an HSI dataset has N pixels, and each pixel has D bands, so we define the HSI dataset as $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N] \in \Re^{D\times N}$, and let $\ell(\mathbf{x}_i) \in \{1, 2, \ldots, c\}$ represent the label information of $\mathbf{x}_i$, where c is the number of class in the HSI. The purpose of the DR method is to obtain the low-dimensional features $\mathbf{Y} = \left[\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N\right] \in \Re^{d\times N}$ of $\mathbf{X}$, where d ($d \ll D$) represents the dimensionality of embedding features. With a projection matrix $\mathbf{P} \in \Re^{D\times d}$, $\mathbf{Y}$ can be expressed as $\mathbf{Y} = \mathbf{P}^T\mathbf{X}$.

A. Mathematical Definition

1) Tangent Space: It is considered as a vector space that is combined by the possible directions in which one can tangentially pass through a point. Intuitively, tangent space is a plane that is tangent to a smooth surface. In a tangent space, a vector that is tangent to a curve on a smooth surface at a given point is termed a tangent vector. Since a smooth surface can be considered as a manifold that can be embedded into a Euclidean space, tangent space can also be represented as an affine space between a point and a Euclidean space that is tangent to the manifold at this point. A tangent space at each point of a manifold has the same dimension as the manifold itself. Fig. 1 shows an example to explain tangent space, where $\mathcal{M}$ is a smooth surface, $T_{\mathcal{M}}$ denotes a tangent plane of $\mathcal{M}$, a curve $\gamma(t)$ is on the smooth surface, $\mathbf{t}_i$ represents a tangent vector at a point v_i on the curve, and $\mathbf{t}_i$ is also a vector on the tangent plane $T_{\mathcal{M}}$.

2) Affine Combination: In a linear space $\mathbf{T} \in \mathbb{F}$, for any N vectors $\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_N$, the affine combination of $\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_N$ is defined as

$$\mathbf{r} = \sum_{i=1}^{N} b_i \mathbf{t}_i \tag{1}$$

where $\sum_{i=1}^{N} b_i = 1, b_i \in \mathbb{F}, i \in \mathbb{N}, 1 \le i \le N$.

An affine space is considered as a vector space without an origin, where the vector only has direction and magnitude. Thus, affine space can be used to describe the invariants of the vector space that is independent of the coordinate system, which is conducive to the study of some invariant properties. The affine combination can be applied to reveal an invariable intrinsic property between a vector space and its transformed affine space.

B. Linear Discriminant Analysis

It is a supervision method [16] to seek an optimal projection space, in which the intraclass data are gathered as much as possible and the interclass data are separated as much as possible.

Based on the intraclass variance and the interclass variance, the optimal projection matrix $\mathbf{P}$ can be obtained by

$$\max \frac{\mathrm{tr}(\mathbf{P}^T\mathbf{S}^b\mathbf{P})}{\mathrm{tr}(\mathbf{P}^T\mathbf{S}^w\mathbf{P})} \tag{2}$$

where $\mathbf{S}^b$ and $\mathbf{S}^w$ denote the between- and within-class scatter matrices, respectively, which are

$$\begin{cases} \mathbf{S}^b = \sum_{i=1}^{c} N_i (\mathbf{u}_i - \mathbf{u})^T (\mathbf{u}_i - \mathbf{u}) \\ \mathbf{S}^w = \sum_{i=1}^{c} \sum_{j=1}^{N_i} \left(\mathbf{x}_j^i - \mathbf{u}_i\right)^T \left(\mathbf{x}_j^i - \mathbf{u}_i\right) \end{cases} \tag{3}$$

where N_i represents the sample size of the ith class, $\mathbf{u}_i$ and $\mathbf{u}$ respectively denote the ith class mean and the total mean, and $\mathbf{x}_j^i$ is the jth sample from the ith class.

C. Graph Embedding

The GE framework defines a graph $\mathbf{G} = \{\mathbf{V}, \mathbf{Q}\}$ to represent a certain statistical or geometric property of data, where $\mathbf{V}$ is the vertex set from X and $\mathbf{Q}$ is a similarity matrix whose elements represent the similarity relationship between vertex pairs [25]. In a low-dimensional space, the similarity that should be maintained is used to reveal the intrinsic homogeneity of data. By preserving the graph structure, the high-dimensional data can be reduced into a low-dimensional space in which the data features possess more robust discriminant performance. Therefore, the GE model is defined as

$$\min_{\mathrm{tr}(\mathbf{YHY}^T)=a} \sum_{i\neq j} \left\|\mathbf{y}_i - \mathbf{y}_j\right\|^2 q_{ij} = \mathbf{YLY}^T \tag{4}$$

where a is a constant, $\mathbf{H}$ is a constraint matrix that is used to avoid a trivial solution of the objective function, q_{ij} is the similarity weight between $\mathbf{x}_i$ and $\mathbf{x}_j$, $\mathbf{L}$ is the Laplacian matrix of graph $\mathbf{G}$, and it can be given as $\mathbf{L} = \mathbf{D} - \mathbf{Q}$ in which $\mathbf{D} = \mathrm{diag}([\sum_j q_{ij}]_i)$ and $\mathbf{Q} = [q_{ij}]_{i,j}$. Generally, $\mathbf{H}$ is defined as an identity matrix or a certain statistical or geometric property of data that should be suppressed.

D. Sparse Representation

It uses [38] a dictionary to represent a sample with a group of representation coefficients, where only few of them are nonzero values and the others are zero values. These nonzero coefficients reveal the intrinsic properties of data. To achieve the sparsity of coefficients, the optimization problem can be constructed as

$$\min_{\mathbf{s}_i} \|\mathbf{x}_i - \mathbf{B}\mathbf{s}_i\|^2 + \lambda \|\mathbf{s}_i\|_1 \tag{5}$$

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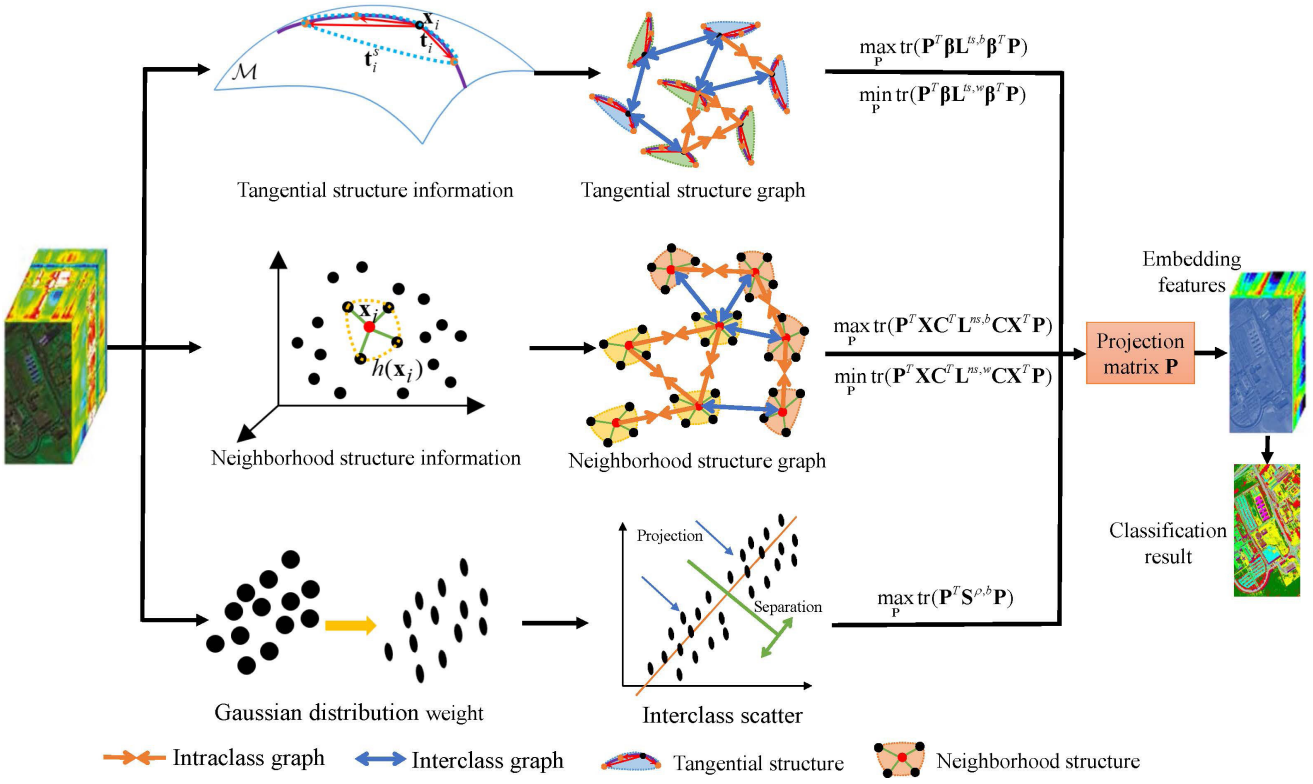


Fig. 2. Flowchart of the proposed MUDE method.

where $\mathbf{B}$ is a given dictionary that is generally defined as $\mathbf{X}$, $\lambda > 0$ is a tradeoff parameter, and $\mathbf{s}_i$ is the sparse coefficients. In this equation, the first term is the reconstruction error, and the second term is the ℓ_1-norm to control the sparsity of coefficients.

III. Multistructure Unified Discriminative Embedding

For the complex intrinsic structures of HSI, only single structure information is very difficult to comprehensively extract the intrinsic features. Therefore, we construct an MUDE method to fuse three embedding models based on the tangential, neighborhood, and statistical characteristics of HSI. We introduce the tangential property of HSI data to obtain the information in tangent space and construct a tangential GE model. Then, we consider the tangential structures of HSI to develop a tangential structure graph and construct an intraclass tangential structure graph and an interclass tangential structure graph. To obtain the neighborhood structures of HSI, we construct an intraclass neighborhood structure graph and an interclass neighborhood structure graph. To further reduce the influence of data distribution for the spectral distortion, we use the statistical information of HSI data to design an interclass Gaussian weighted scatter model. Finally, based on the tangential structure graphs, the neighborhood structure graphs, and the scatter matrix, a feature embedding function is constructed to extract the low-dimensional features, which compacts the homogeneous information and separates the heterogeneous information. The architecture of the proposed method is shown in Fig. 2.

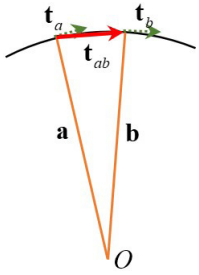


Fig. 3. Example for tangent vector.

A. Tangential Graph

It is used to describe the similar geometric relationship between two samples, which represents the radial direction properties of HSI and reveals the manifold structure in high-dimensional space. However, the vertical direction of each sample also contains some important information of HSI, which represents the tangential properties of HSI and reflects the tangential structure on a manifold in high-dimensional space. In high-dimensional space, the direction of a tangent vector can be approximately considered as the difference of two vectors in HSI data. To better explain this case, we give an example in Fig. 3. In the figure, O is the origin, $\mathbf{a}$ and $\mathbf{b}$ are two normalized vectors, $\mathbf{t}_a$ and $\mathbf{t}_b$ are the tangent vector of $\mathbf{a}$ and $\mathbf{b}$, respectively, and $\mathbf{t}_{ab}$ is equal to $\mathbf{b} - \mathbf{a}$ and it approximately has the same direction with $\mathbf{t}_a$ and $\mathbf{t}_b$ when $\mathbf{a}$ and $\mathbf{b}$ are very close to each other. Thus, we adopt the difference of two vectors to approximately represent the tangent vector. In HSI, vector $\mathbf{a}$ is used to represent the radial direction information, and vector $\mathbf{t}_{ab}$ is used to denote the tangential information.

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For the HSI dataset $\mathbf{X}$, we construct a tangential graph $\mathbf{G}^t = \{\mathbf{V}^t, \mathbf{Q}^t\}$, where $\mathbf{V}^t$ is the vertex set that is composed of the tangent vectors of $\mathbf{X}$ and $\mathbf{Q}^t$ is the weight matrix that reveals the similarity relationship between two vertices. To reduce the influence of magnitude, we normalize the tangent vectors to construct the tangential graph. Here, we use the difference between each sample and its nearest neighbor (NN) sample to approximately represent the tangent vector. In a low-dimensional space, the tangential property of HSI is preserved to obtain the projection matrix $\mathbf{P}$, and the embedding function is defined as

$$\min_{\mathrm{tr}(\mathbf{P}^T\mathbf{T}\mathbf{T}^T\mathbf{P})=1} \frac{1}{2}\sum_{i\neq j}\left\|\mathbf{P}^T\mathbf{t}_i - \mathbf{P}^T\mathbf{t}_j\right\|^2 q_{ij}^t = \frac{1}{2}\sum_{i\neq j}\left\|\mathbf{P}^T\frac{\mathbb{N}_1(\mathbf{x}_i)-\mathbf{x}_i}{\|\mathbb{N}_1(\mathbf{x}_i)-\mathbf{x}_i\|} - \mathbf{P}^T\frac{\mathbb{N}_1(\mathbf{x}_j)-\mathbf{x}_j}{\left\|\mathbb{N}_1(\mathbf{x}_j)-\mathbf{x}_j\right\|}\right\|^2 q_{ij}^t \quad (6)$$

where $\mathbb{N}_1(\mathbf{x}_i)$ denotes the NN of $\mathbf{x}_i$ and $\mathbf{t}_i = (\mathbb{N}_1(\mathbf{x}_i) - \mathbf{x}_i / \|\mathbb{N}_1(\mathbf{x}_i) - \mathbf{x}_i\|)$ is the normalized tangent vector that is the difference between $\mathbb{N}_1(\mathbf{x}_i)$ and $\mathbf{x}_i$, q_{ij}^t reflects the similarity weight between the tangent vectors of $\mathbf{x}_i$ and $\mathbf{x}_j$, and it is defined as

$$q_{ij}^t = \begin{cases} \exp\left(-\dfrac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{2\delta_i^2}\right), & \text{if } \mathbf{x}_j \in \mathbb{N}_k(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where $\delta_i = (1/k)\sum_j \|\mathbf{x}_i - \mathbf{x}_j\|$, k is the number of neighbors, and $\mathbb{N}_k(\mathbf{x}_i)$ denotes the k NNs of $\mathbf{x}_i$.

With some mathematical transformation, we can get the function as

$$\min_{\mathrm{tr}(\mathbf{V}^T\mathbf{T}\mathbf{T}^T\mathbf{V})=1} \mathrm{tr}(\mathbf{P}^T\mathbf{T}\mathbf{L}^t\mathbf{T}^T\mathbf{P}) \quad (8)$$

where $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_N]$, $\mathbf{L}^t = \mathbf{D}^t - \mathbf{Q}^t$ is the tangential Laplacian matrix, $\mathbf{Q}^t = [q_{ij}^t]_{i,j}$ is the weight matrix, and $\mathbf{D}^t = \mathrm{diag}([\sum_j q_{ij}^t]_i)$ is a diagonal matrix that reveals the degrees of vertex.

B. Tangential Structure Graph

In the tangential graph, it just considers one tangent vector of each sample, which is very difficult to comprehensively reveal the intrinsic information for the complex HSI. Thus, we consider the tangential structure information of each sample to construct a tangential structure graph, where we use the affine combination of the normalized tangent vectors to represent the tangential structure information. Therefore, for a sample $\mathbf{x}_i$, we select its k NNs to construct the tangent vectors, and its tangential structure can be denoted as

$$\mathbf{t}_i^s = \sum_{\mathbf{x}_j\in\mathbb{N}_k(\mathbf{x}_i)} \frac{\mathbf{x}_j - \mathbf{x}_i}{\left\|\mathbf{x}_j - \mathbf{x}_i\right\|} s_{ij}^{ts} \quad (9)$$

where $\mathbf{t}_i^s$ is the tangential structure representation of $\mathbf{x}_i$, $\mathbb{N}_k(\mathbf{x}_i)$ denotes the k NNs of $\mathbf{x}_i$, and s_{ij}^{ts} is the affine combination coefficients of the tangent vector between $\mathbf{x}_i$ and its neighbor $\mathbf{x}_j$ and $\sum_{\mathbf{x}_j\in\mathbb{N}_k(\mathbf{x}_i)} s_{ij}^{ts} = 1$.

In the tangential structure representation, it needs to first give the neighborhood size and the affine combination coefficients, which is very difficult to set the appropriate values. To obtain good tangential structure representation, according to the thought of SR, we use the affine combination of all the tangent vectors between each sample and other samples and constrain the coefficients with ℓ_1-norm to adaptively represent the tangential structure information. For the coefficients, they are sparse and only few of them are nonzero values. Since the coefficients are sparse, a few tangent vectors at a point are used for the affine combination that reflects the tangential properties of data. The objective function can be defined as

$$\min_{\mathbf{s}_i^{ts}} \frac{1}{2}\left\|\sum_j \frac{\mathbf{x}_j - \mathbf{x}_i}{\left\|\mathbf{x}_j - \mathbf{x}_i\right\|} s_{ij}^{ts}\right\|^2 + \lambda\sum_j s_{ij}^{ts}. \quad (10)$$

To match the affine constraint $\sum_j s_{ij}^{ts} = 1$, we add a weight for s_{ij}^{ts} as the ℓ_1-norm term, and the objective function can be redefined as

$$\begin{cases} \min\limits_{\mathbf{s}_i^{ts}} \dfrac{1}{2}\left\|\mathbf{T}_i^s\mathbf{s}_i^{ts}\right\|^2 + \lambda\left\|\mathrm{diag}(\mathbf{F}_i^s)\mathbf{s}_i^{ts}\right\|_1 \\ \text{s.t. } \mathbf{1}^T\mathbf{s}_i^{ts} = 1 \end{cases} \quad (11)$$

where $\mathbf{s}_i^{ts} = [s_{ij}^{ts}]_j$ is the tangential sparse coefficients to reveal the similarity relationship between the tangent vectors and $\lambda > 0$ is the tradeoff parameter between the affine combination and the sparse constraint. The structure tangent vectors $\mathbf{T}_i^s$ at $\mathbf{x}_i$ and the weight $\mathbf{F}_i^s$ are denoted as

$$\mathbf{T}_i^s = \left[\frac{\mathbf{x}_j - \mathbf{x}_i}{\left\|\mathbf{x}_j - \mathbf{x}_i\right\|}\right]_{j,j\neq i} \in \Re^{D\times(N-1)} \quad (12)$$

$$\mathbf{F}_i^s = \left[\frac{\left\|\mathbf{x}_j - \mathbf{x}_i\right\|}{\sum_{k\neq i}\|\mathbf{x}_k - \mathbf{x}_i\|}\right]_{j,j\neq i} \in \Re^{1\times(N-1)}. \quad (13)$$

To reduce the influence of heterogeneous information, we just use the intraclass samples in the objective function as the following:

$$\begin{cases} \min\limits_{\mathbf{s}_i^{ts,w}} \dfrac{1}{2}\left\|\mathbf{T}_i^{s,w}\mathbf{s}_i^{ts,w}\right\|^2 + \lambda\left\|\mathrm{diag}(\mathbf{F}_i^{s,w})\mathbf{s}_i^{ts,w}\right\|_1 \\ \text{s.t. } \mathbf{1}^T\mathbf{s}_i^{ts,w} = 1 \end{cases} \quad (14)$$

where $\mathbf{s}_i^{ts,w}$ is the intraclass tangential sparse coefficients and $\mathbf{T}_i^{s,w}$ and $\mathbf{F}_i^{s,w}$ are obtained from the same class samples and they are denoted as

$$\mathbf{T}_i^{s,w} = \left\{\mathbf{t}_{ik}^{s,w} | \mathbf{t}_{ik}^{s,w} \in \mathbf{T}_i^s, \ell(\mathbf{x}_i) = \ell(\mathbf{x}_k)\right\} \quad (15)$$

$$\mathbf{F}_i^{s,w} = \left\{\mathbf{f}_{ik}^{s} | \mathbf{f}_{ik}^{s} \in \mathbf{F}_i^s, \ell(\mathbf{x}_i) = \ell(\mathbf{x}_k)\right\} \quad (16)$$

where $\mathbf{t}_{ik}^{s,w}$ is the tangent vector between $\mathbf{x}_i$ and the kth sample from the same class.

With the alternating direction method of multipliers (ADMM), we can obtain the intraclass tangential sparse coefficients $\mathbf{s}_i^{ts,w}$ at $\mathbf{x}_i$. Then, we can define the intraclass tangential structure representation coefficients $\mathbf{A}^{ts,w} = [a_{ij}^{ts,w}]_{i,j} \in \Re^{N\times(N-1)}$ as the following:

$$a_{ij}^{t,w} = \begin{cases} s_{ik}^{t,w}, & \text{if } s_{ik}^{ts,w} \neq 0 \text{ and } j = \Gamma\left(\mathbf{t}_{ik}^{s,w}, \mathbf{T}_i^s\right) \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

where $\Gamma(\mathbf{t}_{ik}^{s,w}, \mathbf{T}_i^s)$ is an indicator function that is used to get the index of the tangent vector $\mathbf{t}_{ik}^{s,w}$ in the structure tangent vectors $\mathbf{T}_i^s$.

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According to the intraclass sparse coefficients $\mathbf{a}_i^{ts,w} = [a_{ij}^{ts,w}]_j$, we can obtain the tangential structure information at $\mathbf{x}_i$ as $f(\mathbf{x}_i) = \mathbf{T}_i^s(\mathbf{a}_i^{ts,w})^T$. Then, we construct a tangential structure graph $\mathbf{G}^{ts} = \{\mathbf{V}^{ts}, \mathbf{Q}^{ts}\}$, where $\mathbf{V}^{ts}$ denotes the vertex set that consists of the tangential structure representation $f(\mathbf{x}_i)$ and $\mathbf{Q}^{ts} = [q_{ij}^{ts}]_{i,j}$ is the weight of edges to represent the similarity between two tangential structure vectors. We should preserve the similarity relationships to extract the low-dimensional features that reduce the redundant information and enhance the discriminating power for the classification task. The embedding objective function can be denoted as

$$\begin{aligned}\min_{\mathbf{P}} \frac{1}{2}\sum_{i\neq j}\left\|\mathbf{P}^T f(\mathbf{x}_i) - \mathbf{P}^T f(\mathbf{x}_j)\right\|^2 q_{ij}^{ts} &= \frac{1}{2}\sum_{i\neq j}\left\|\mathbf{P}^T\mathbf{T}_i^s\left(\mathbf{a}_i^{t,w}\right)^T - \mathbf{P}^T\mathbf{T}_j^s\left(\mathbf{a}_j^{t,w}\right)^T\right\|^2 q_{ij}^{ts} \\ &= \frac{1}{2}\sum_{i\neq j}\left\|\mathbf{P}^T\mathbf{r}_i - \mathbf{P}^T\mathbf{r}_j\right\|^2 q_{ij}^{ts} \\ &= \mathrm{tr}(\mathbf{P}^T\mathbf{R}\mathbf{L}^{ts}\mathbf{R}^T\mathbf{P}) \end{aligned} \tag{18}$$

where $\mathbf{R} = [\mathbf{r}_i]_i$, $\mathbf{r}_i = \mathbf{T}_i^s(\mathbf{a}_i^{t,w})^T$, $\mathbf{L}^{ts} = \mathbf{D}^{ts} - \mathbf{Q}^{ts}$ is the tangential structure Laplacian matrix, and $\mathbf{D}^{ts} = \mathrm{diag}([\sum_j q_{ij}^{ts}]_i)$ is a diagonal degree matrix of vertex.

To enhance the discriminating performance of embedding features, we make full use of the category information to divide the tangential structure graph into two graphs, including an intraclass tangential structure graph and an interclass tangential structure graph, which will be beneficial to compact the samples from the same class and separate the samples from different classes. For the intraclass tangential structure graph $\mathbf{G}^{ts,w} = \{\mathbf{V}^{ts}, \mathbf{Q}^{ts,w}\}$, we adopt the intraclass sparse coefficients to adaptively define its connected relationship and the similarity weight. If the coefficient between two vertices is zero, we should not connect the two vertices and the weight is set to zero; otherwise, the two vertices are connected and given a weight as the following:

$$q_{ij}^{ts,w} = \begin{cases} \exp\left(-\dfrac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2 + \left(a_{ij}^{ts,w}\right)^2}{2\left(\delta_i^{d,w}\right)^2 + 2\left(\delta_i^{a,w}\right)^2}\right), & \text{if } a_{ij}^{ts,w} \neq 0 \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

where $\delta_i^{d,w} = (1/\|\mathbf{a}_i^{ts,w}\|_0)\sum_j |a_{ij}^{ts,w}| \times \|\mathbf{x}_i - \mathbf{x}_j\|$ and $\delta_i^{a,w} = (1/\|\mathbf{a}_i^{ts,w}\|_0)\sum_{a_{ij}^{ts,w}\neq 0} a_{ij}^{ts,w}$.

Then, an intraclass tangential structure embedding function is constructed to compact the homogeneous information of data as follows:

$$\min_{\mathbf{P}} \frac{1}{2}\sum_{i\neq j}\left\|\mathbf{P}^T\mathbf{T}_i^s\left(\mathbf{a}_i^{ts,w}\right)^T - \mathbf{P}^T\mathbf{T}_j^s\left(\mathbf{a}_j^{ts,w}\right)^T\right\|^2 q_{ij}^{ts,w} = \mathrm{tr}\left(\mathbf{P}^T\mathbf{R}\mathbf{L}^{ts,w}\mathbf{R}^T\mathbf{P}\right) \tag{20}$$

where $\mathbf{L}^{ts,w} = \mathbf{D}^{ts,w} - \mathbf{Q}^{ts,w}$ is the intraclass tangential structure Laplacian matrix, $\mathbf{D}^{ts,w} = \mathrm{diag}([\sum_j q_{ij}^{ts,w}]_i)$ is a diagonal degree matrix of vertex, and $\mathbf{Q}^{ts,w} = [q_{ij}^{ts,w}]_{i,j}$ is the weight matrix.

To construct the interclass tangential structure graph $\mathbf{G}^{ts,b} = \{\mathbf{V}^{ts}, \mathbf{Q}^{ts,b}\}$, it first needs to obtain the connected relationships of the tangent vectors that is the differences between a sample and the samples from different classes. We also use the sparse affine representation to adaptively obtain the intrinsic relationship between the tangent vectors, and the objective function is defined as

$$\begin{cases} \min\limits_{\mathbf{s}_i^{ts,b}} \dfrac{1}{2}\left\|\mathbf{T}_i^{s,b}\mathbf{s}_i^{ts,b}\right\|^2 + \lambda\left\|\mathrm{diag}\left(\mathbf{F}_i^{s,b}\right)\mathbf{s}_i^{ts,b}\right\|_1 \\ \text{s.t.}\mathbf{1}^T\mathbf{s}_i^{ts,b} = 1 \end{cases} \tag{21}$$

where $\mathbf{s}_i^{ts,b}$ is the interclass tangential sparse coefficients and $\mathbf{T}_i^{s,b}$ and $\mathbf{F}_i^{s,b}$ are obtained from the samples belonging to different classes, which are defined as

$$\mathbf{T}_i^{s,b} = \left\{\mathbf{t}_{ik}^{s,b} | \mathbf{t}_{ik}^{s,b} \in \mathbf{T}_i^s, \ell(\mathbf{x}_i) \neq \ell(\mathbf{x}_k)\right\} \tag{22}$$

$$\mathbf{F}_i^{s,b} = \left\{\mathbf{f}_{ik}^{s} | \mathbf{f}_{ik}^{s} \in \mathbf{F}_i^s, \ell(\mathbf{x}_i) \neq \ell(\mathbf{x}_k)\right\} \tag{23}$$

where $\mathbf{t}_{ik}^{s,b}$ is the tangent vector between $\mathbf{x}_i$ and the kth sample from different classes.

With the ADMM, we can obtain the interclass tangential structure representation coefficients $\mathbf{A}^{ts,b} = [a_{ij}^{ts,b}]_{i,j} \in \Re^{N\times(N-1)}$ as follows:

$$a_{ij}^{t,b} = \begin{cases} s_{ik}^{t,b}, & \text{if } s_{ik}^{ts,b} \neq 0 \text{ and } j = \Gamma(\mathbf{t}_{ik}^{s,b}, \mathbf{T}_i^s) \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

Then, we can obtain the weight of the interclass tangential structure graph as

$$q_{ij}^{ts,b} = \begin{cases} \exp\left(-\dfrac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2 + \left(a_{ij}^{ts,b}\right)^2}{2\left(\delta_i^{d,b}\right)^2 + 2\left(\delta_i^{a,b}\right)^2}\right), & \text{if } a_{ij}^{ts,b} \neq 0 \\ 0, & \text{otherwise} \end{cases} \tag{25}$$

where $\delta_i^{d,b} = (1/\|\mathbf{a}_i^{ts,b}\|_0)\sum_j |a_{ij}^{ts,b}| \times \|\mathbf{x}_i - \mathbf{x}_j\|$ and $\delta_i^{a,b} = (1/\|\mathbf{a}_i^{ts,b}\|_0)\sum_{a_{ij}^{ts,b}\neq 0} a_{ij}^{ts,b}$, $\mathbf{a}_i^{ts,b} = [a_{ij}^{ts,b}]_j$.

According to the weight $\mathbf{Q}^{ts,b} = [q_{ij}^{ts,b}]_{i,j}$, we use the tangential structure information obtaining from the same class samples to construct an interclass embedding function, which aims to separate the similar samples from different classes

$$\max_{\mathbf{P}} \frac{1}{2}\sum_{i\neq j}\left\|\mathbf{P}^T\mathbf{T}_i^s\left(\mathbf{a}_i^{ts,w}\right)^T - \mathbf{P}^T\mathbf{T}_j^s\left(\mathbf{a}_j^{ts,w}\right)^T\right\|^2 q_{ij}^{ts,b} = \mathrm{tr}(\mathbf{P}^T\mathbf{R}\mathbf{L}^{ts,b}\mathbf{R}^T\mathbf{P}) \tag{26}$$

where $\mathbf{L}^{ts,b} = \mathbf{D}^{ts,b} - \mathbf{Q}^{ts,b}$ is the interclass tangential structure Laplacian matrix and $\mathbf{D}^{ts,b} = \mathrm{diag}([\sum_j q_{ij}^{ts,b}]_i)$ is a diagonal matrix.

C. Neighborhood Structure Graph

In HSI, the neighborhood samples should have the similar spectra, while these spectra have a large difference in real application, even if they are from the same class due to the influence of imaging condition. Therefore, only using the individual information of each sample to construct DR model, it is very difficult to reveal intrinsic properties of HSI. To better

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extract the intrinsic feature of HSI, we consider the neighborhood structure to construct a graph $\mathbf{G}^{ns} = \{\mathbf{V}^{ns}, \mathbf{Q}^{ns}\}$, where the vertex set $\mathbf{V}^{ns}$ is composed of the neighborhood structure of each sample. Here, we adopt the local linear reconstruction to obtain the neighborhood structure $h(\mathbf{x}_i)$. To better obtain the neighborhood structure, we select k intraclass NNs to represent the neighborhood structure as

$$h(\mathbf{x}_i) = \sum_{\mathbf{x}_j \in \mathbb{N}_{k,w}(\mathbf{x}_i)} c_{ij}\mathbf{x}_j = \sum_{j=1}^{n} c_{ij}\mathbf{x}_j = \mathbf{X}\mathbf{c}_i^T \tag{27}$$

where $\mathbb{N}_{k,w}(\mathbf{x}_i)$ is the k NNs of $\mathbf{x}_i$ in the same class, c_{ij} is the structure representation coefficients between $\mathbf{x}_i$ and $\mathbf{x}_j$, c_{ij} is set to zero for the nonneighbors, and $\mathbf{c}_i = [c_{ij}]_j$.

To get the neighborhood structure, we minimize the reconstruction errors to obtain the structure representation coefficients, and the objective function is defined as

$$\begin{cases} \min \sum_{i=1}^{n} \|\mathbf{x}_i - h(\mathbf{x}_i)\|^2 = \sum_{i=1}^{n} \left\| \mathbf{x}_i - \sum_{j=1}^{n} c_{ij}\mathbf{x}_j \right\|^2 \\ \text{s.t.} \sum_{j=1}^{n} c_{ij} = 1. \end{cases} \tag{28}$$

Then, we can get

$$c_{ij} = \begin{cases} \dfrac{\sum_{m=1}^{k} \left(z_{jm}^{i}\right)^{-1}}{\sum_{p=1}^{k}\sum_{q=1}^{k} \left(z_{pq}^{i}\right)^{-1}}, & \text{if } \mathbf{x}_j \in \mathbb{N}_{k,w}(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{29}$$

where $\mathbf{z}_i = [z_{mn}^{i}]_{k\times k} = [\mathbf{x}_i - \mathbf{x}_i^1, \ldots, \mathbf{x}_i - \mathbf{x}_i^k]^T[\mathbf{x}_i - \mathbf{x}_i^1, \ldots, \mathbf{x}_i - \mathbf{x}_i^k]$ and $\mathbf{x}_i^j$ is the jth NN of $\mathbf{x}_i$.

According to the class label, we construct an intraclass neighborhood graph $\mathbf{G}^{ns,w} = \{\mathbf{V}^{ns}, \mathbf{Q}^{ns,w}\}$ and an interclass neighborhood graph $\mathbf{G}^{ns,b} = \{\mathbf{V}^{ns}, \mathbf{Q}^{ns,b}\}$ to compact the samples from the same class and separate the samples from different classes. The intraclass weight and the interclass weight can be represented as the following:

$$q_{ij}^{ns,w} = \begin{cases} \exp\left(-\dfrac{\left\|h(\mathbf{x}_i) - h(\mathbf{x}_j)\right\|^2}{2\left(\delta_i^{ns,w}\right)^2}\right), & \text{if } \mathbf{x}_j \in \mathbb{N}_{k,w}(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{30}$$

$$q_{ij}^{ns,b} = \begin{cases} \exp\left(-\dfrac{\left\|h(\mathbf{x}_i) - h(\mathbf{x}_j)\right\|^2}{2\left(\delta_i^{ns,b}\right)^2}\right), & \text{if } \mathbf{x}_j \in \mathbb{N}_{k,b}(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{31}$$

where $\delta_i^{ns,w} = (1/k)\sum_{\mathbf{x}_j \in \mathbb{N}_{k,w}(\mathbf{x}_i)} \|h(\mathbf{x}_i) - h(\mathbf{x}_j)\|, \delta_i^{ns,b} = (1/k)\sum_{\mathbf{x}_j \in \mathbb{N}_{k,b}(\mathbf{x}_i)} \|h(\mathbf{x}_i) - h(\mathbf{x}_j)\|$ and $\mathbb{N}_{k,w}(\mathbf{x}_i)$ and $\mathbb{N}_{k,b}(\mathbf{x}_i)$ are the intraclass and interclass k NNs of $\mathbf{x}_i$, respectively.

Based on the two neighborhood structure graphs, we design two objective functions to embed the high-dimensional data into a low-dimensional space, which aims to reduce the distance between the intraclass samples and enlarge the distance between the interclass samples. The obtained features have better intraclass compactness and interclass separability

$$\begin{aligned} &\min_{\mathbf{P}} \frac{1}{2}\sum_{i,j} \left\|\mathbf{P}^T h(\mathbf{x}_i) - \mathbf{P}^T h(\mathbf{x}_j)\right\|^2 q_{ij}^{ns,w} \\ &\quad = \frac{1}{2}\sum_{i,j} \left\|\mathbf{P}^T\mathbf{X}\mathbf{c}_i^T - \mathbf{P}^T\mathbf{X}\mathbf{c}_i^T\right\|^2 q_{ij}^{ns,w} \end{aligned} \tag{32}$$

$$\begin{aligned} &\max_{\mathbf{P}} \frac{1}{2}\sum_{i,j} \left\|\mathbf{P}^T h(\mathbf{x}_i) - \mathbf{P}^T h(\mathbf{x}_j)\right\|^2 q_{ij}^{ns,b} \\ &\quad = \frac{1}{2}\sum_{i,j} \left\|\mathbf{P}^T\mathbf{X}\mathbf{c}_i^T - \mathbf{P}^T\mathbf{X}\mathbf{c}_i^T\right\|^2 q_{ij}^{ns,b}. \end{aligned} \tag{33}$$

With some mathematical transformation, the two functions can be represented as

$$\begin{cases} \min_{\mathbf{P}} \operatorname{tr}(\mathbf{P}^T\mathbf{X}\mathbf{C}^T\mathbf{L}^{ns,w}\mathbf{C}\mathbf{X}^T\mathbf{P}) \\ \max_{\mathbf{P}} \operatorname{tr}(\mathbf{P}^T\mathbf{X}\mathbf{C}^T\mathbf{L}^{ns,b}\mathbf{C}\mathbf{X}^T\mathbf{P}) \end{cases} \tag{34}$$

where $\mathbf{L}^{ns,w} = \mathbf{D}^{ns,w} - \mathbf{Q}^{ns,w}$ and $\mathbf{L}^{ns,b} = \mathbf{D}^{ns,b} - \mathbf{Q}^{ns,b}$ are the intraclass and interclass neighborhood structure Laplacian matrices, $\mathbf{D}^{ns,w} = \operatorname{diag}([\sum_j q_{ij}^{ns,w}]_i)$, $\mathbf{D}^{ns,b} = \operatorname{diag}([\sum_j q_{ij}^{ns,b}]_i)$, $\mathbf{Q}^{ns,w} = [q_{ij}^{ns,w}]_{i,j}$, $\mathbf{Q}^{ns,b} = [q_{ij}^{ns,b}]_{i,j}$, and $\mathbf{C} = [\mathbf{c}_i]_i$ is the neighborhood structure coefficient matrix.

D. Interclass Gaussian Weighted Scatter Model

Since the complex imaging environment of HSI results in the distribution difference of data, to further reduce the influence of data distribution for spectral distortion and enhance the representation performance of features, we consider the statistical properties of HSI and design an interclass Gaussian weighted scatter model. The Gaussian distribution model is defined as

$$\widehat{\mathbf{e}}_i = \left[\frac{1}{\sqrt{2\pi\sigma_j^2}}\exp\left(-\frac{(x_{ij} - u_j)^2}{2\sigma_j^2}\right)\right]_{j=1}^{D} \tag{35}$$

where $\widehat{\mathbf{e}}_i$ is the Gaussian distribution of $\mathbf{x}_i = [x_{ij}]_{j=1}^{D}$ and $\sigma_j^2 = (1/N)\sum_{i=1}^{N}(x_{ij} - u_j)^2$ and $u_j = (1/N)\sum_{i=1}^{N} x_{ij}$ are the variance and the mean value at the jth dimension, respectively.

With the Gaussian distribution, we define a weight for each dimension of each sample to reduce the influence of spectral distortion. To unify the scale factor, the weight is represented as

$$\mathbf{e}_i = \frac{\widehat{\mathbf{e}}_i}{\left\|\widehat{\mathbf{e}}_i\right\|_1}. \tag{36}$$

In a low-dimensional space, we consider the distribution of data to construct an interclass Gaussian weighted scatter model, which improves the distribution difference of data and enhances the discriminating performance of the low-dimensional features from different classes. We design an objective function based on the interclass variance as follows:

$$\begin{aligned} &\max_{\mathbf{P}} \sum_{j=1}^{c} N_j \left\| \mathbf{P}^T \frac{\sum_{i=1}^{N_j} \mathbf{e}_i^j \odot \mathbf{x}_i^j}{N_j} - \mathbf{P}^T \frac{\sum_{k=1}^{c}\sum_{l=1}^{N_j} \mathbf{e}_l^k \odot \mathbf{x}_l^k}{N} \right\|^2 \\ &\qquad = \operatorname{tr}(\mathbf{P}^T\mathbf{S}^{e,b}\mathbf{P}) \end{aligned} \tag{37}$$

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where $\odot$ denotes the element-by-element multiplication to control the distribution of each dimension, N_j is the number of the jth class samples, $\mathbf{x}_i^j$ is the ith sample from the jth class, $\mathbf{e}_i^j$ is the Gaussian distribution weight of $\mathbf{x}_i^j$, and $\mathbf{S}^{e,b}$ is the interclass Gaussian weighted scatter matrix and is defined as

$$\mathbf{S}^{e,b} = \sum_{j=1}^{c} N_j \left(\begin{array}{c} \frac{\sum_{i=1}^{N_j} \mathbf{e}_i^j \odot \mathbf{x}_i^j}{N_j} - \\ \frac{\sum_{k=1}^{c} \sum_{l=1}^{N_j} \mathbf{e}_l^k \odot \mathbf{x}_l^k}{N} \end{array} \right)^T \left(\begin{array}{c} \frac{\sum_{i=1}^{N_j} \mathbf{e}_i^j \odot \mathbf{x}_i^j}{N_j} - \\ \frac{\sum_{k=1}^{c} \sum_{l=1}^{N_j} \mathbf{e}_l^k \odot \mathbf{x}_l^k}{N} \end{array} \right). \quad (38)$$

Therefore, for the tangential, neighborhood, and statistical properties of HSI, we can obtain

$$\begin{cases} \max\limits_{\mathbf{P}} \operatorname{tr}(\mathbf{P}^T \mathbf{R} \mathbf{L}^{ts,b} \mathbf{R}^T \mathbf{P}) \\ \min\limits_{\mathbf{P}} \operatorname{tr}(\mathbf{P}^T \mathbf{R} \mathbf{L}^{ts,w} \mathbf{R}^T \mathbf{P}) \\ \max\limits_{\mathbf{P}} \operatorname{tr}(\mathbf{P}^T \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,b} \mathbf{C} \mathbf{X}^T \mathbf{P}) \\ \min\limits_{\mathbf{P}} \operatorname{tr}(\mathbf{P}^T \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,w} \mathbf{C} \mathbf{X}^T \mathbf{P}) \\ \max\limits_{\mathbf{P}} \operatorname{tr}(\mathbf{P}^T \mathbf{S}^{e,b} \mathbf{P}). \end{cases} \quad (39)$$

E. Feature Embedding

To better reveal the intrinsic features of HSI, we use the tangential, neighborhood, and statistical properties of HSI to construct a multistructure complementary model. Based on the two tangential structure graphs, two neighborhood structure graphs, and the interclass Gaussian weighted scatter model, we compact the intraclass information and separate the interclass information as much as possible in a low-dimensional space, and the feature embedding function is defined as

$$\max J(\mathbf{P}) = \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}} \left(\begin{array}{l} (1-\eta) \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,b} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,b} \mathbf{R}^T \end{array} \right) \\ +\eta \mathbf{S}^{e,b} \end{array} \right) \mathbf{P} \right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}} \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,w} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,w} \mathbf{R}^T \end{array} \right) \mathbf{P} \right)} \quad (40)$$

where $0 < \alpha < 1$ and $0 < \eta < 1$ are the tradeoff parameters.

To solve the function, it can be transformed into the following form:

$$\begin{cases} \max\limits_{\mathbf{P}} \mathbf{P}^{\mathrm{T}} \left(\begin{array}{l} (1-\eta) \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,b} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,b} \mathbf{R}^T \end{array} \right) \\ +\eta \mathbf{S}^{e,b} \end{array} \right) \mathbf{P} \\ \text{s.t. } \mathbf{P}^{\mathrm{T}} \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,w} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,w} \mathbf{R}^T \end{array} \right) \mathbf{P} = \mathbf{I}. \end{cases} \quad (41)$$

We can use the Lagrangian multiplier ξ to construct the Lagrangian function as follows:

$$L(\mathbf{P}, \lambda) = \mathbf{P}^{\mathrm{T}} \left((1-\eta) \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,b} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,b} \mathbf{R}^T \end{array} \right) + \eta \mathbf{S}^{e,b} \right) \mathbf{P} + \xi \left(\mathbf{P}^{\mathrm{T}} \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,w} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,w} \mathbf{R}^T \end{array} \right) \mathbf{P} - \mathbf{I} \right). \quad (42)$$

Let the derivative of $\mathbf{P}$ be zero to get the optimal values

$$\left((1-\eta) \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,b} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,b} \mathbf{R}^T \end{array} \right) + \eta \mathbf{S}^{e,b} \right) \mathbf{p}_i = \xi_i \left(\begin{array}{l} \alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,w} \mathbf{C} \mathbf{X}^T \\ +(1-\alpha) \mathbf{R} \mathbf{L}^{ts,w} \mathbf{R}^T \end{array} \right) \mathbf{p}_i \quad (43)$$

where $\mathbf{p}_i$ is the eigenvector and ξ_i is the corresponding eigenvalue. We can obtain the projection matrix $\mathbf{P} = \left[\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_d \right] \in \Re^{D \times d}$ with the eigenvectors of d largest eigenvalues, which integrates the structure information and three characteristics to project the high-dimensional data into a low-dimensional space. Thus, the low-dimensional features $\mathbf{Y}$ of data $\mathbf{X}$ can be represented by

$$\mathbf{Y} = \mathbf{P}^T \mathbf{X} \in \Re^{d \times N}. \quad (44)$$

MUDE uses the intraclass neighborhood and tangential structure information to compact the samples from the same class, and the intraclass scatter, neighborhood, and tangential structure information to enhance the diversity of the samples from different classes, which is beneficial to the subsequent classification task. Its steps are given in Algorithm 1.

According to Algorithm 1, we can obtain the computational complexity of the proposed method. For the intraclass and interclass tangential sparse coefficients, both need to solve by ADMM with the complexity of $O(N^4)$. Both take $O(N^2)$ to obtain the weights of the intraclass and interclass tangential structure graphs. It requires $O(Nk^2D)$ to obtain the neighborhood structure representation coefficients. In the two neighborhood graphs, the complexity of the intraclass and interclass weights is $O(N^2)$. The cost is $O(ND^2)$ to compute the interclass Gaussian weighted scatter matrix. The matrices $\alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,b} \mathbf{C} \mathbf{X}^T + (1-\alpha) \mathbf{R} \mathbf{L}^{ts,b} \mathbf{R}^T$ and $\alpha \mathbf{X} \mathbf{C}^T \mathbf{L}^{ns,w} \mathbf{C} \mathbf{X}^T + (1-\alpha) \mathbf{R} \mathbf{L}^{ts,w} \mathbf{R}^T$, and their computational complexity is both $O(N^2 D)$. To solve the generalized eigenvalue problem, it costs $O(D^3)$ to obtain the projection matrix. Therefore, the total computational complexity is $O(N^4 + Nk^2D + ND^2 + N^2D + D^3)$, which mainly depends on the neighborhood size, the sample size, and the number of bands. Generally, the neighborhood size is a small value and the sample size is larger than the number of bands, and thus, the complexity is about $O(N^4)$, which indicates that the solution of tangential sparse coefficients takes a lot of time.

IV. Experimental Results and Discussion

In this section, we will verify MUDE on three hyperspectral datasets and compare with some of the state-of-the-art DR algorithms.

A. Datasets

1) PaviaU Dataset: It is an HSI located in the northern part of the University of Pavia in northern Italy, acquired by the reflective optics system imaging spectrometer (ROSIS) sensor. The dataset has a size of 610 × 340 pixels, a total of 115 spectral bands, and a spatial resolution of 1.3 m. After the 12 bands affected by water and gas, the remaining 103 bands are used for experimental research, mainly including nine types of ground objects, such as soil, trees, and bricks. The false-color images and real features are shown in Fig. 4.

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Algorithm 1 MUDE

Input: Samples $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N] \in \Re^{D\times N}$ and corresponding class label $\ell(\mathbf{x}_i) \in \{1, 2, \cdots, c\}$, embedding tradeoff parameters α and η, neighborhood size k, sparse tradeoff parameter λ, and embedding dimension d.

Output: Low-dimensional embedding features $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N] \in \Re^{d\times N}$.

1: Compute the intraclass tangential structure coefficients $\mathbf{a}_i^{ts,w} = [a_{ij}^{ts,w}]_j$ by adaptive sparse representation based affine combination.

2: Compute the tangential structure information $\mathbf{r} = [\mathbf{r}_i]_i$, where $\mathbf{r}_i = \mathbf{T}_i^s(\mathbf{a}_i^{t,w})^T$.

3: Construct the intraclass tangential structure graph $\mathbf{G}^{ts,w} = \{\mathbf{V}^{ts}, \mathbf{Q}^{ts,w}\}$, where the vertices are composed of $\mathbf{R}$ and the weight $q_{ij}^{ts,w}$ is defined by $a_{ij}^{ts,w}$.

4: Compute the interclass tangential structure coefficients $\mathbf{a}_i^{ts,b} = [a_{ij}^{ts,b}]_j$ by adaptive sparse representation based affine combination.

5: Construct the interclass tangential structure graph $\mathbf{G}^{ts,b} = \{\mathbf{V}^{ts}, \mathbf{Q}^{ts,b}\}$, where the vertices are composed of $\mathbf{R}$ and the weight $q_{ij}^{ts,b}$ is defined by $a_{ij}^{ts,b}$.

6: Calculate the intraclass and interclass tangential structure Laplacian matrices $\mathbf{L}^{ts,w} = \mathbf{D}^{ts,w} - \mathbf{Q}^{ts,w}$ and $\mathbf{L}^{ts,b} = \mathbf{D}^{ts,b} - \mathbf{Q}^{ts,b}$.

7: Compute the neighborhood structure coefficients $\mathbf{C} = [\mathbf{c}_i]_i$ using k intraclass nearest neighbors.

8: Construct the intraclass neighborhood graph $\mathbf{G}^{ns} = \{\mathbf{V}^{ns}, \mathbf{Q}^{ns,w}\}$, where $\mathbf{Q}^{ns,w} = [q_{ij}^{ns,w}]_{i,j}$ is defined by k intraclass nearest neighbors.

9: Construct the interclass neighborhood graph $\mathbf{G}^{ns} = \{\mathbf{V}^{ns}, \mathbf{Q}^{ns,b}\}$, where $\mathbf{Q}^{ns,w} = [q_{ij}^{ns,w}]_{i,j}$ is defined by k interclass nearest neighbors.

10: Calculate the intraclass and interclass neighborhood structure Laplacian matrices $\mathbf{L}^{ns,w} = \mathbf{D}^{ns,w} - \mathbf{Q}^{ns,w}$ and $\mathbf{L}^{ns,b} = \mathbf{D}^{ns,b} - \mathbf{Q}^{ns,b}$.

11: Compute the interclass Gaussian weighted scatter matrix $\mathbf{S}^{e,b}$.

12: Solve the generalized eigenvalue problem:

$$\left((1-\eta)\begin{pmatrix}\alpha\mathbf{X}\mathbf{C}^T\mathbf{L}^{ns,b}\mathbf{C}\mathbf{X}^T\\+(1-\alpha)\mathbf{R}\mathbf{L}^{ts,b}\mathbf{R}^T\end{pmatrix}+\eta\mathbf{S}^{e,b}\right)\mathbf{p}_i = \xi_i\begin{pmatrix}\alpha\mathbf{X}\mathbf{C}^T\mathbf{L}^{ns,w}\mathbf{C}\mathbf{X}^T\\+(1-\alpha)\mathbf{R}\mathbf{L}^{ts,w}\mathbf{R}^T\end{pmatrix}\mathbf{p}_i.$$

13: Obtain the projection matrix with d largest eigenvalues corresponding eigenvectors:
$\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_d] \in \Re^{D\times d}$.

14: Calculate the low-dimensional embedding features:
$\mathbf{Y} = \mathbf{P}^T\mathbf{X} \in \Re^{d\times N}$.

2) HoustonU Dataset: It is an HSI of the University of Houston campus and its adjacent urban areas acquired by the ITRES-CASI 1500 hyperspectral imager. The size of the dataset is 349 × 1905 pixels, the spatial resolution is 2.5 m, and there are 144 bands in total, and It includes 15 types of ground objects, such as grass, soil, and road. The false-color image and real ground object image are shown in Fig. 5.

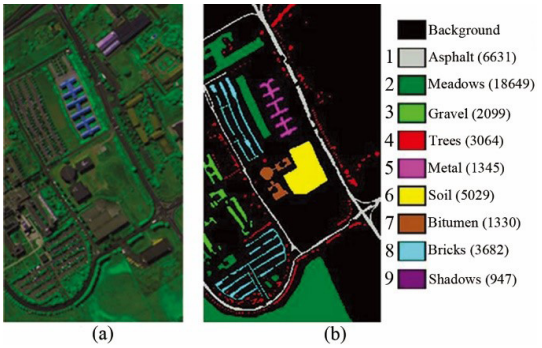


Fig. 4. PaviaU HSI. (a) HSI in false color. (b) Corresponding ground truth. (Note that the number of samples for each class is shown in brackets.)

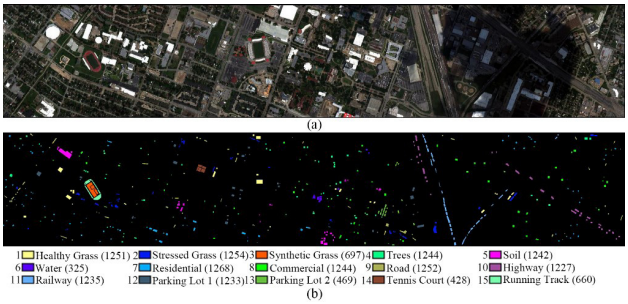


Fig. 5. HoustonU HSI. (a) HSI in false color. (b) Corresponding ground truth. (Note that the number of samples for each class is shown in brackets.)

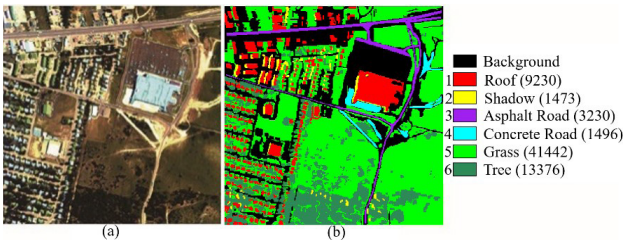


Fig. 6. Urban HSI. (a) HSI in false color. (b) Corresponding ground truth. (Note that the number of samples for each class is shown in brackets).

3) Urban Dataset: It is an HSI of the area near Fort Hood, TX, USA, captured in 1995 by the HYDICE sensor. The image size is 307 × 307 pixels, and the spatial resolution is 4 m. There are 210 bands in total, and 48 are removed. Among the bands affected by noise, the remaining 162 bands are used for experiments. The image contains six types of features: roof, shadow, asphalt road, concrete road, grass, and tree. The false-color images and real features are shown in Fig. 6.

B. Experimental Setup

In the experiment, a certain number of labeled samples were randomly selected for training, and the rest samples were used for testing. The training samples were learned through each DR algorithm to obtain a low-dimensional projection matrix, then, the test samples were embedded into the low-dimensional projection space, and finally, a classifier was selected to discriminate the class labels of test samples. We used the average accuracy (AA) of all classes, the overall classification accuracy (OA), and the kappa coefficient (KC) as evaluation indicators. In this article, we compared MUDE with

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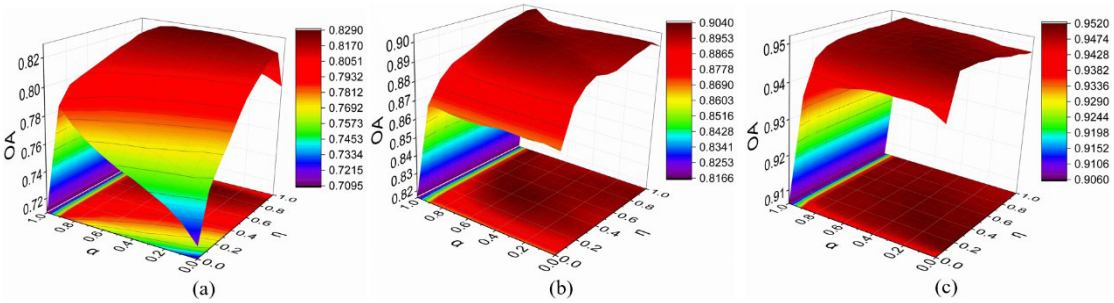


Fig. 7. OAs with respect to parameters α and η on (a) PaviaU, (b) HoustonU, and (c) Urban datasets.

several of the latest DR algorithms. To analyze the tangential representation, we chose LCSMHL, SDME, and LMSDL as a comparison. We used LGSFA and LFDA to analyze the local information. To compare the adaptive GE, we selected several representation learning algorithms, including DLSP, CGDA, SGDA, and SLGDA. We also used the raw spectral (RAW) of the HSI data as the baseline for comparison. Then, the NN classifier and the spectral angle mapper (SAM) were used to classify the low-dimensional features obtained by different DR algorithms.

In the experiment, we randomly selected 20, 30, 40, 50, and 60 samples from each class on the PaviaU, HoustonU, and Urban datasets for training, and the rest as test samples. In order to obtain good experimental results, we used cross validation to optimize the parameters of each method. For all the methods on each dataset, the embedding dimension was set to 25. For MUDE, we set the tradeoff parameters and from 0 to 1 at the interval of 0.1 to find the optimal tradeoff parameter and used 1 as the interval to find the optimal number of neighborhood sizes in the range from 1 to 25. The sparse tradeoff parameter was set from 0.01 to 20 at the interval of 1 to find the optimal value. Each experiment was repeated ten times, and the mean with standard variance (STD) was taken as the experimental result. All experiments were performed using MATLAB 2018b on a computer with Gold-6136 CPU, 32-GB RAM, and 64-bit Windows 10.

C. Parameter Analysis

In MUDE, there are four important parameters, including embedding tradeoff parameters α and η, neighborhood size k, and sparse tradeoff parameter λ. In this section, we analyzed the change of OAs under different parameter settings. We randomly selected 40 samples from each class on the three datasets as a training set and used NN for classification. For each condition, we performed ten repeated experiments to obtain the average classification accuracies.

1) *Embedding Tradeoff Parameters α and η:* α is used to balance the neighborhood information and the tangential information, and η is used to control the contribution of interclass Gaussian weighted scatter. Fig. 7 shows the change of OAs under different values of α and η. For the three datasets, OAs both improve with the increase of α and η and then begin to become a downward trend when exceeding a certain threshold because different models can obtain different features. Only when the parameters are set to a proper balanced value, the proposed models can perform a better complementary relationship. When α or η is set to 0 or 1, MUDE will degenerate into one of neighborhood structure graph embedding (NSGE), tangential structure graph embedding (TSGE), enhanced neighborhood structure embedding (ENSGE), enhanced TSGE (ETSGE), or multistructure graph embedding (MSGE), which achieves lower OAs than MUDE. This case indicates that the proposed models can perform the complementarity between different intrinsic properties of HSI data and improve the discriminating performance of embedding features. The parameters α and η are set to 0.9 and 0.7 for the PaviaU dataset, 0.5 and 0.5 for the HoustonU dataset, and 0.3 and 0.3 for the Urban dataset.

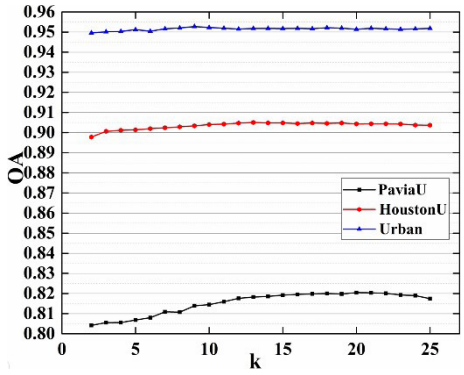


Fig. 8. OAs with respect to neighborhood size k.

2) *Neighborhood Size* k*:* k is used to represent k NNs of each sample. As shown in Fig. 8, we can find that OAs first increase and then decrease with the change of neighborhood size on the three datasets. When k is small, there are not enough neighborhood samples to reveal the structure relationship of HSI data. On the contrary, when k is large, some unrelated samples will be introduced to result in a negative effect. In experiments, we set k to 20 for the PaviaU dataset, 13 for the HoustonU dataset, and 10 for the Urban dataset.
3) *Sparse Tradeoff Parameter λ:* λ is used to control the sparsity of the affine combination coefficients. Fig. 9 shows the results with different values of λ. In this figure, OAs first increase and then decrease with the increase of λ. Only using a proper value of λ, we can achieve the affine representation to better reveal the

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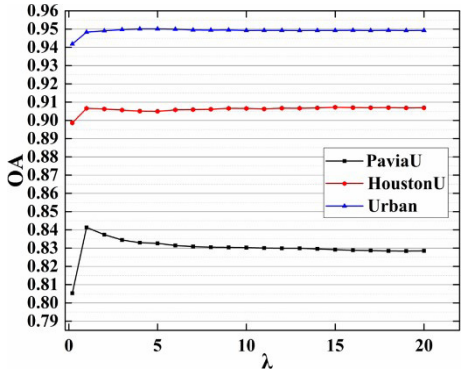


Fig. 9. OAs with respect to sparse tradeoff parameter λ.

tangential properties of HSI data. The parameter λ is set to 1 for the PaviaU and HoustonU datasets and 5 for the Urban dataset.

D. Ablation Study

To verify the effectiveness of the proposed modules, we construct the following models and conduct comparative experiments on the three datasets. In this study, we randomly selected 40 samples from each class as a training set and used NN for classification. We show the average OAs with STDs and KCs of ten repeated experiments.

1) *NSGE:* In order to maintain the geometric structure of the homogeneous region, we propose an NSGE method, which uses the neighborhood structure information to perform the GE, and its objective function is

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,b}\mathbf{C}\mathbf{X}^{T}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,w}\mathbf{C}\mathbf{X}^{T}\mathbf{P}\right)}. \tag{45}$$

2) *Tangential Graph Embedding (TGE):* To reveal the tangential properties of data, we propose the TGE method using the tangent vectors represented by each sample and its NN. The objective function is defined as

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{T}\mathbf{L}^{t,b}\mathbf{T}^{T}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{T}\mathbf{L}^{t,w}\mathbf{T}^{T}\mathbf{P}\right)}. \tag{46}$$

3) *TSGE:* To better reveal the intrinsic tangential characteristics of the complex structure information in hyperspectral data, we propose a TSGE method. The objective function is defined as

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{R}\mathbf{L}^{ts,b}\mathbf{R}^{T}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{R}\mathbf{L}^{ts,w}\mathbf{R}^{T}\mathbf{P}\right)}. \tag{47}$$

4) *Gaussian Weighted Discriminant Analysis (GWDA):* To verify the effectiveness of interclass Gaussian weighted scatter, we propose a GWDA method to compare with LDA, and its objective function is

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{S}^{e,b}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{S}^{w}\mathbf{P}\right)}. \tag{48}$$

5) *ENSGE:* Based on the NSGE method, we add an interclass Gaussian weighted scatter model and propose an ENSGE method. The objective function is defined as

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\begin{pmatrix}(1-\eta)\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,b}\mathbf{C}\mathbf{X}^{T}\\ +\eta\mathbf{S}^{e,b}\end{pmatrix}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,w}\mathbf{C}\mathbf{X}^{T}\mathbf{P}\right)}. \tag{49}$$

6) *ETSGE:* Based on the TSGE method, we add an interclass Gaussian weighted scatter model and propose an ETSGE method. The objective function is

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\begin{pmatrix}(1-\eta)\mathbf{R}\mathbf{L}^{ts,b}\mathbf{R}^{T}\\ +\eta\mathbf{S}^{e,b}\end{pmatrix}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\mathbf{R}\mathbf{L}^{ts,b}\mathbf{R}^{T}\mathbf{P}\right)}. \tag{50}$$

7) *MSGE:* We construct a complementary GE model by combining NSGE and TSGE and propose an MSGE method with an objective function as follows:

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\begin{pmatrix}\alpha\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,b}\mathbf{C}\mathbf{X}^{T}\\ +(1-\alpha)\mathbf{R}\mathbf{L}^{ts,b}\mathbf{R}^{T}\end{pmatrix}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\begin{pmatrix}\alpha\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,w}\mathbf{C}\mathbf{X}^{T}\\ +(1-\alpha)\mathbf{R}\mathbf{L}^{ts,w}\mathbf{R}^{T}\end{pmatrix}\mathbf{P}\right)}. \tag{51}$$

8) *Discriminative MSGE (DMSGE):* To compare MUDE and verify the effectiveness of the interclass Gaussian weighted scatter, we use the between-class scatter to replace the interclass Gaussian weighted scatter and propose an MSGE method, and its objective function is as follows:

$$\max_{\mathbf{P}} \frac{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\begin{pmatrix}(1-\eta)\begin{pmatrix}\alpha\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,b}\mathbf{C}\mathbf{X}^{T}\\ +(1-\alpha)\mathbf{R}\mathbf{L}^{ts,b}\mathbf{R}^{T}\end{pmatrix}\\ +\eta\mathbf{S}^{b}\end{pmatrix}\mathbf{P}\right)}{\operatorname{tr}\left(\mathbf{P}^{\mathrm{T}}\begin{pmatrix}\alpha\mathbf{X}\mathbf{C}^{T}\mathbf{L}^{ns,w}\mathbf{C}\mathbf{X}^{T}\\ +(1-\alpha)\mathbf{R}\mathbf{L}^{ts,w}\mathbf{R}^{T}\end{pmatrix}\mathbf{P}\right)}. \tag{52}$$

TABLE I

CLASSIFICATION RESULTS WITH DIFFERENT MODELS ON THE THREE DATASETS (OA ± STD (KC) %)

Method	Data sets		
	PaviaU	HoustonU	Urban
MFA	71.5±2.0 (64.1)	83.6±2.4 (82.3)	92.3±1.2 (90.6)
NSGE	71.8±2.7 (64.3)	87.0±0.7 (86.0)	93.7±0.8 (92.3)
ENSGE	75.8±2.2 (69.2)	87.0±0.9 (85.9)	93.9±0.8 (92.5)
TGE	76.0±1.9 (69.5)	84.4±1.1 (83.1)	92.8±0.7 (91.2)
TSGE	77.7±1.7 (71.4)	88.9±1.0 (88.0)	94.0±0.7 (93.4)
ETSGE	80.9±1.4 (75.5)	89.1±0.9 (88.2)	94.1±0.7 (93.4)
MSGE	78.8±2.3 (73.0)	89.1±0.7 (88.4)	94.0±0.9 (93.5)
LDA	69.5±1.6 (61.6)	73.3±1.3 (71.1)	91.1±1.0 (89.1)
GWDA	70.2±1.8 (63.0)	73.8±1.2 (71.7)	92.8±1.0 (91.2)
DMSGE	82.0±1.7 (76.8)	89.1±0.7 (88.2)	94.2±0.8 (93.0)
MUDE	**82.7**±1.9 (**77.7**)	**90.0**±0.9 (**88.9**)	**94.7**±0.7 (**93.7**)

For the three datasets (PaviaU, HoustonU, and Urban), the parameter η in the ENSGE method is set to {0.5, 0.5, 0.4}, the parameter η in the ETSGE method is set to {0.7, 0.5, 0.3}, the parameter α in the MSGE method is set to {0.5, 0.4, 0.5}, and the parameters α and η in the DMSGE method are set to {0.9, 0.5, 0.3} and {0.7, 0.5, 0.3}, respectively. Table I gives the results for different models on the three datasets.

In Table I, we can see that NSGE and TSGE, respectively, have better results than MFA and TGE, which indicates that the structure information of each sample is more effective to

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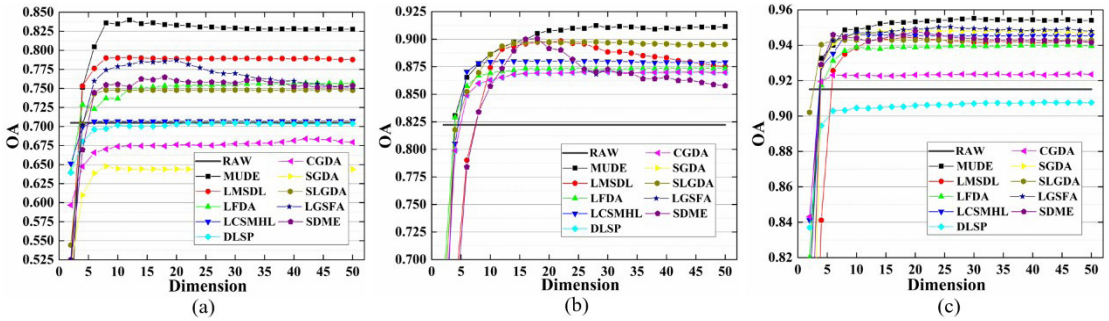


Fig. 10. OAs with respect to embedding dimension d on (a) PaviaU, (b) HoustonU, and (c) Urban datasets.

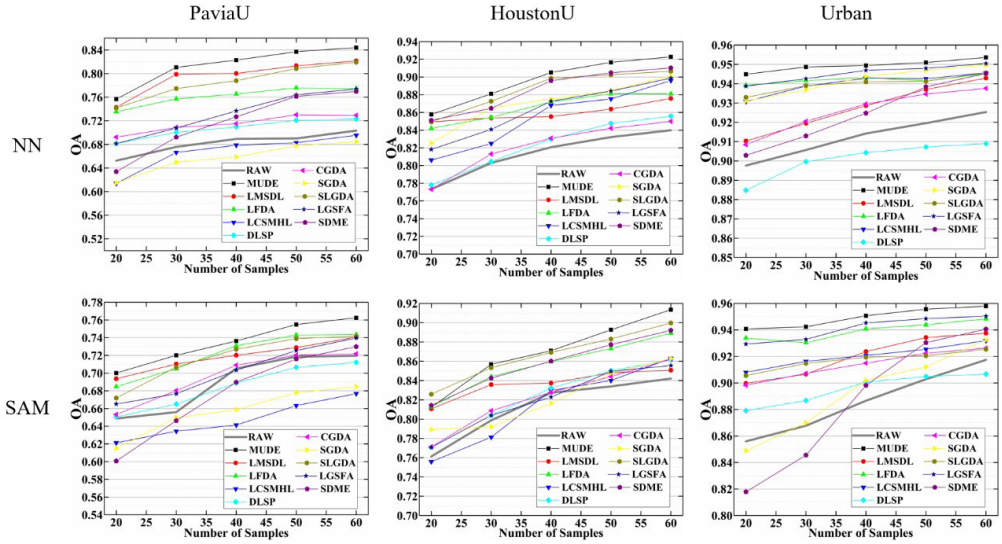


Fig. 11. OAs with different numbers of samples and classifiers on the three datasets.

describe the intrinsic complex geometric distribution relationship of HSI data compared with the individual information of each sample. ENSGE and ETSGE have better classification accuracy than NSGE and TSGE because ENSGE and ETSGE also consider the statistical properties, which enhances the separability of samples from different classes. Compared with NSGE and TSGE, the classification results of MSGE are both better because MSGE uses the neighborhood and tangential structures to achieve the complementarity of different intrinsic structure information and obtain more discriminative features. At the same time, comparing GWDA with LDA and MUDE with DMSGE, we can find that the introduction of interclass gaussian weighted scatter can make the sample more separable. In general, MUDE obtains the best classification results since it integrates the excellent properties of multiple structures and obtains more discriminative features. In the low-dimensional space, MUDE can promote the clustering of intraclass samples and the separability of interclass samples.

E. Embedding Dimension Analysis

Since different embedding dimensions will affect the effectiveness of the DR algorithms, we conducted some experiments on the three datasets with different embedding dimensions. Here, we randomly selected 40 samples from each class for training and used NN for classification. We set the range of dimension from 2 to 50 with the interval of 2, and the results are shown in Fig. 10.

In this figure, with the increase of d, the accuracies of most algorithms rise to a peak because a large embedding dimension contains more discriminative information. When the embedding dimension d continues to grow, the valuable information will reach saturation, which will make the OAs no longer increase to maintain a peak value or slightly decrease. In addition, most methods have better results than RAW because these methods can effectively reduce the redundant information of data. For a few methods (especially for LGSFA and SDME) on some datasets, the OAs begin to decrease when exceeding a certain dimension, which is termed the Hughes phenomenon. For all the methods, MUDE is obviously superior to other methods in most cases, which indicates that the proposed method can extract more discriminative features with the complementarity of the neighborhood, tangential, and statistical properties of HSI data.

F. Classification Results and Running Time

To compare the classification performance of each algorithm, we randomly selected 20, 30, 40, 50, and 60 samples from each class as the training set, and the remaining samples were used as the test set. To demonstrate the robustness for different classifiers, we used NN and SAM for classification.

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TABLE II
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PAVIAU DATASET (%)

Class	RAW	LFDA	LCSMHL	LGSFA	CGDA	SGDA	SLGDA	DLSP	SDME	LMSDL	MUDE
1	70.9	73.3	65.8	64.0	70.9	71.9	74.9	70.8	65.3	72.2	**79.7**
2	63.4	70.3	66.4	74.1	63.3	63.3	**82.5**	63.7	72.4	79.5	81.7
3	70.7	73.6	62.6	70.8	70.9	64.8	79.6	70.7	72.5	76.3	**81.1**
4	96.8	97.1	95.5	95.3	96.9	84.1	94.3	**97.0**	96.4	91.4	95.5
5	99.1	99.4	99.5	**99.8**	99.1	99.0	**99.8**	99.1	99.6	98.8	**99.8**
6	69.0	74.4	71.0	**84.4**	69.0	71.8	83.1	68.8	80.8	81.2	84.2
7	89.6	91.7	84.6	87.4	89.5	88.2	89.5	88.1	89.1	91.4	**92.1**
8	67.9	70.1	61.0	56.4	68.0	66.6	63.1	69.2	58.2	64.8	**71.3**
9	99.8	99.9	99.9	99.9	99.8	99.8	99.4	99.8	99.8	99.5	**100.0**
AA	80.8	83.3	78.5	81.4	80.8	78.9	85.2	80.8	81.6	83.9	**87.3**
OA	71.0	75.4	70.5	75.3	71.0	70.1	81.5	71.1	74.7	79.4	**83.0**
KC	63.7	68.9	63.0	68.7	63.6	62.6	76.1	63.9	68.0	73.5	**78.1**
Time (s)	-	0.0	3.1	0.4	0.0	0.1	89.5	0.8	2.1	2.2	6.8

TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTONU DATASET (%)

Class	RAW	LFDA	LCSMHL	LGSFA	CGDA	SGDA	SLGDA	DLSP	SDME	LMSDL	MUDE
1	98.0	98.2	96.0	96.5	98.0	97.2	96.0	96.8	97.8	95.1	**98.2**
2	**97.8**	97.6	97.7	96.6	**97.8**	95.6	96.1	95.9	97.0	89.5	96.4
3	97.8	99.5	**100.0**	**100.0**	97.8	99.8	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
4	97.0	97.0	97.5	97.0	97.0	97.0	**98.1**	90.5	95.0	95.2	96.6
5	96.7	97.7	98.2	99.3	96.7	97.0	99.5	99.0	**99.7**	**99.7**	99.3
6	98.9	99.2	99.6	99.6	98.9	99.6	99.6	99.2	**100.0**	99.6	99.6
7	85.0	87.9	89.7	88.8	84.9	86.8	87.1	77.8	**93.2**	85.3	92.2
8	71.5	76.1	**88.4**	84.6	71.5	79.7	86.2	77.8	84.7	81.9	86.9
9	75.6	79.9	78.3	72.7	75.3	77.1	78.9	72.8	80.5	73.3	**83.3**
10	76.9	84.2	84.4	82.6	76.9	89.3	90.3	91.7	92.1	80.7	**94.6**
11	72.7	80.0	83.9	80.5	72.7	81.2	84.2	79.0	82.0	69.3	**88.1**
12	72.0	82.4	82.6	74.6	72.0	85.5	81.7	74.2	79.6	60.7	**89.7**
13	40.3	47.2	49.6	**78.7**	40.3	53.8	61.9	45.5	66.7	72.6	61.9
14	92.7	99.2	99.7	99.5	92.9	98.4	**100.0**	98.1	99.5	99.7	99.7
15	98.5	98.5	98.7	**98.8**	98.5	97.8	**98.8**	98.5	**98.8**	**98.8**	**98.8**
AA	84.8	88.3	89.6	90.0	84.8	89.1	90.6	86.5	91.1	86.8	**92.4**
OA	84.8	88.4	89.8	88.7	84.8	89.0	90.3	86.2	90.7	85.0	**92.6**
KC	83.5	87.4	89.0	87.8	83.5	88.1	89.5	85.0	90.0	83.8	**92.0**
Time (s)	-	0.0	26.3	1.1	0.1	0.2	185.4	2.3	1.9	9.8	11.0

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE URBAN DATASET (%)

Class	RAW	LFDA	LCSMHL	LGSFA	CGDA	SGDA	SLGDA	DLSP	SDME	LMSDL	MUDE
1	83.0	85.8	86.1	85.7	84.3	**89.8**	86.9	82.2	87.4	86.2	88.5
2	92.1	92.3	90.8	94.0	91.0	92.7	88.7	92.0	93.9	93.6	**94.1**
3	96.2	96.9	97.0	**97.1**	96.5	96.2	96.7	92.8	96.7	95.8	96.8
4	94.6	96.2	96.1	96.3	95.0	95.0	96.7	90.9	94.0	93.8	**97.3**
5	95.8	**98.5**	98.1	98.4	95.8	98.1	96.2	93.6	98.3	98.2	**98.5**
6	89.6	91.9	93.0	94.0	89.7	93.9	93.2	90.2	92.8	93.0	**94.1**
AA	91.9	93.6	93.5	94.1	92.1	94.2	93.0	90.3	93.9	93.4	**94.9**
OA	92.2	94.0	93.9	94.3	92.5	94.4	93.5	90.4	94.2	93.7	**95.1**
KC	90.5	92.6	92.6	93.1	90.8	93.3	92.0	88.4	92.9	92.3	**93.9**
Time (s)	-	0.0	6.8	0.3	0.0	0.1	26.4	0.5	2.6	1.4	4.5

Fig. 11 shows the average OAs with ten repeated experiments under each condition.

In Fig. 11, for all the methods, the OAs both improve with the increase of training samples as more priori samples can be used to train a good model and enhance the representation performance of model. The proposed MUDE method can achieve the best classification accuracies than the other methods with different classifiers under different numbers of samples. This is because MUDE considers different information to construct a complementary model and possesses stronger robust performance to match different classifiers. In addition, MUDE considers the structure information of each sample to describe the complex intrinsic structure of HSI, while the other methods just use the individual information of each sample.

To analyze the results for each class, we used 60 samples from each class for training and NN for classification to display the accuracies of each class and running time on the PaviaU, HoustonU, and Urban datasets in Tables II–IV.

According to Tables II–IV, the proposed method achieves better classification results in most classes than the other methods. Meanwhile, MUDE possesses the beset OAs, AAs, and KCs in this condition because MUDE considers not only the structure information of each sample to describe the

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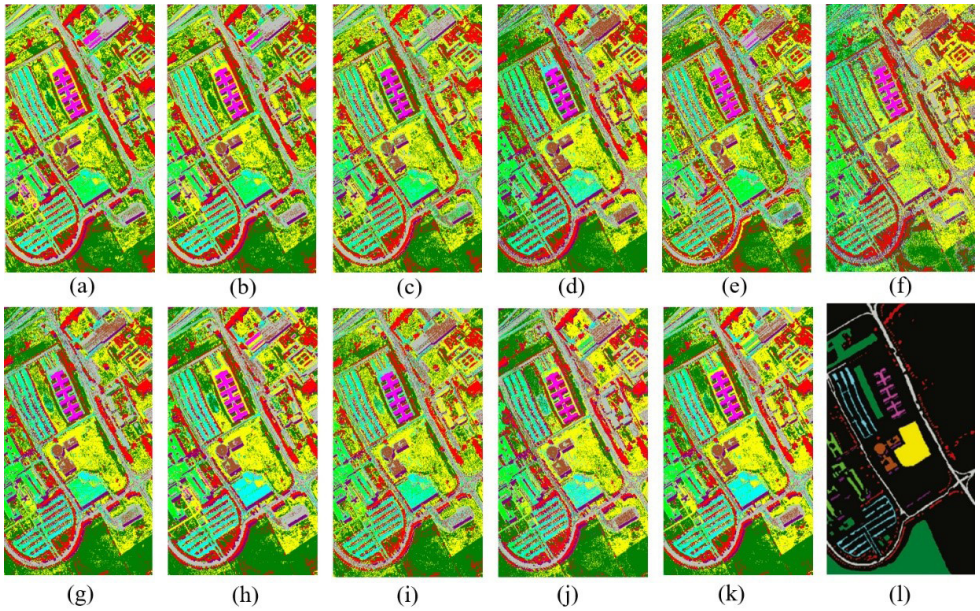


Fig. 12. Classification maps of different methods on the PaviaU dataset. (a) RAW. (b) LFDA. (c) LCSMHL. (d) LGSFA. (e) CGDA. (f) SGDA. (g) SLGDA. (h) DLSP. (i) SDME. (j) LMSDL. (k) MUDE. (l) Ground truth.

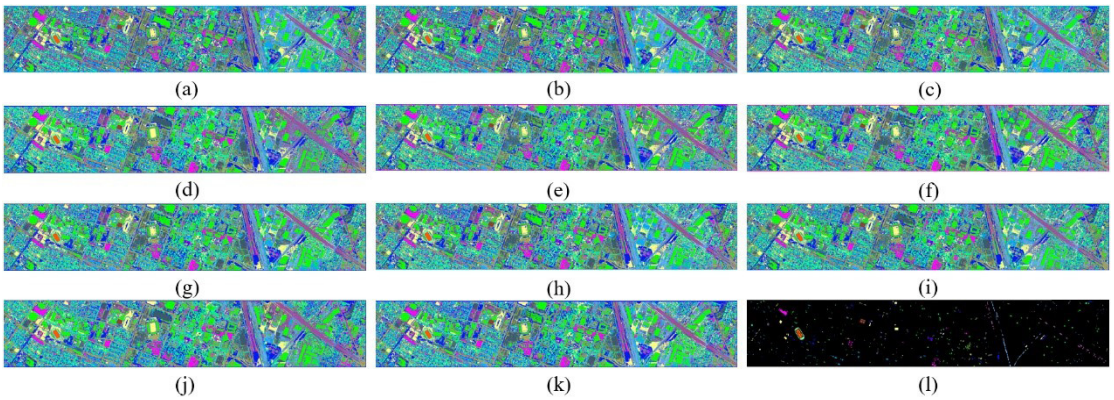


Fig. 13. Classification maps of different methods on the HoustonU dataset. (a) RAW. (b) LFDA. (c) LCSMHL. (d) LGSFA. (e) CGDA. (f) SGDA. (g) SLGDA. (h) DLSP. (i) SDME. (j) LMSDL. (k) MUDE. (l) Ground truth.

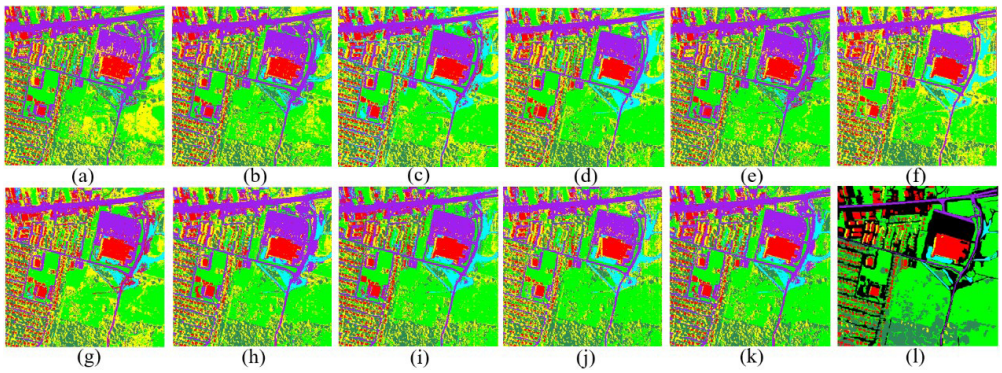


Fig. 14. Classification maps of different methods on the Urban dataset. (a) RAW. (b) LFDA. (c) LCSMHL. (d) LGSFA. (e) CGDA. (f) SGDA. (g) SLGDA. (h) DLSP. (i) SDME. (j) LMSDL. (k) MUDE. (l) Ground truth.

complex HSI but also different properties (including neighborhood, tangential, and statistical characteristics) to perform the complementarity of multiview information of HSI data. The classification results indicate that the proposed method can better represent the intrinsic properties of HSI data and extract more discriminative features to improve the classification performance.

For the running time, MUDE takes more time to learn the model than the other methods except for SLGDA because MUDE needs to construct the neighborhood structure graphs,

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the tangential structure graph, and the interclass Gaussian weighted scatter model and extract structure information of each sample, while the other methods just use a certain characteristic and individual information of each sample to design the learning models. For SLGDA, it needs to solve the sparse and low-rank representation with very high computational complexity, and thus, it takes more running time to obtain the model than MUDE.

To visualize the classification results for the whole dataset, we presented the corresponding classification maps of Tables II–IV in Figs. 12–14.

According to Figs. 12–14, on the three datasets, it can be seen that MUDE generates more homogeneous regions and obtains smoother classification maps than the other methods, especially in the areas of *Asphalt*, *Gravel*, *Metal*, and *Bricks* for the PaviaU dataset, *Road*, *Highway*, *Railway*, and *Parking Lot 1* for the HoustonU dataset and *Shadow*, *Concrete Road*, and *Tree* for the Urban dataset. The proposed method uses multiple complex structures of each sample to represent the intrinsic information of HSI data and makes full use of multiple discriminative features to compact the intraclass samples and separate the interclass samples.

V. Conclusion

In this article, we propose a DR method, termed MUDE, to extract the intrinsic features of HSI for classification. In this method, we use the tangential structure information obtained by the adaptive affine combination in tangent space to design two tangential structure graphs to represent the intraclass and the interclass tangential properties of HSI. Furthermore, we construct two neighborhood structure graphs by using the local intraclass neighborhood reconstruction structure of each sample to complement the tangential structure information. In addition, to reduce the influence of the data distribution, we develop an interclass Gaussian weighted scatter model to better improve the difference between different classes and enhance the discriminating power of embedding features. Based on these structure models to reveal the neighborhood, tangential, and statistical properties of HSI, we construct a feature embedding function to compact the homogeneous information from the same class samples and separate the heterogeneous information from different class samples. For the proposed MUDE model, it not only considers the structure information of each sample but also makes full use of the different views information to better represent the intrinsic characteristics of HSI. From the experimental results on the PaivaU, HoustonU, and Urban HSI datasets, MUDE can effectively improve the classification performance of HSI compared with other related algorithms. In the future work, we will use the advantages of HSI spatial information to obtain more efficient feature extraction methods.

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